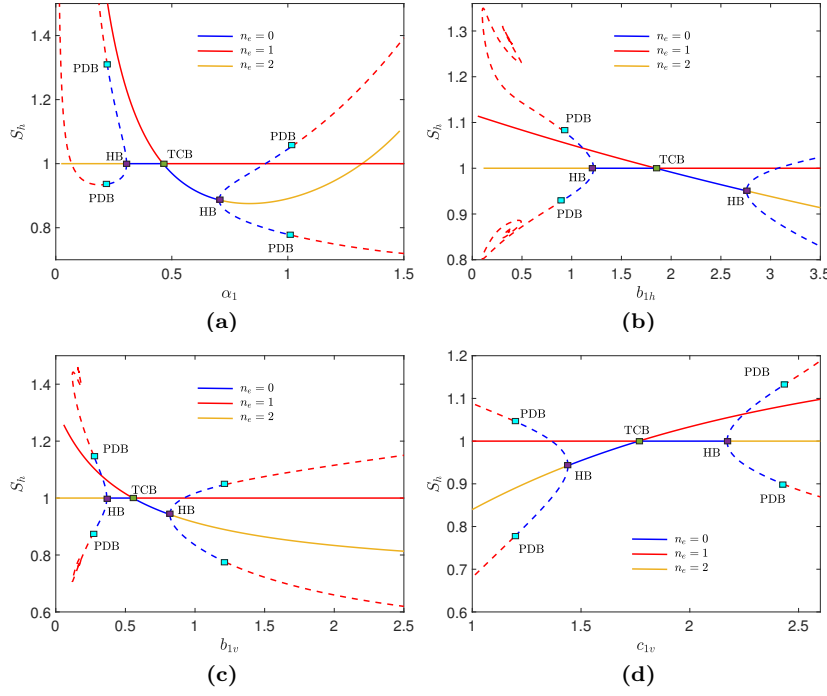




**Fig. 5** Single parameter numerical continuation of susceptible human population with respect to parameters  $\alpha_1$  (a),  $b_{1h}$  (b),  $b_{1v}$  (c) and  $c_{1v}$  (d) with  $\tau_h = \tau_v = \tau = 0$  and  $\tau_r = 2.4$ . The colored lines represent the number of eigenvalues with positive real parts. The dashed lines stand for the branches emanating from Hopf bifurcation and represent the maximum of the periodic orbits. Purple, cyan and green squares respectively represent Hopf, Period doubling and transcritical bifurcations.

#### 5.4 Disease free equilibrium (DFE)

Disease free equilibrium is found as  $\Sigma_2^* = \left(1, 0, 0, 0, \frac{1}{c_{1v}}, 0, 0\right)$ . Using the linearization argument the explicit form of the matrix given in Eq. (25) is given as follows

$$J_\tau = \begin{pmatrix} -1 & 0 & 0 & \omega & 0 & 0 & -fe^{-\lambda\tau_h} \\ 0 & -\sigma_h & 0 & 0 & 0 & 0 & \alpha_1 fe^{-\lambda\tau_h} \\ 0 & 1 & -\xi_h e^{-\lambda\tau_r} & 0 & 0 & 0 & 0 \\ 0 & 0 & e^{-\lambda\tau_r} & -\eta_h & 0 & 0 & 0 \\ 0 & 0 & -he^{-\lambda\tau_v} & 0 & -c_{1v} & 0 & 0 \\ 0 & 0 & \alpha_2 he^{-\lambda\tau_v} & 0 & 0 & -c_{2v} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -c_{1v} \end{pmatrix}.$$

The calculations performed in Sec. 5.1 are valid here with the exception of Eq. (31). Since only susceptible human and susceptible vector population exist for